

Name: Prajwal Tanjal Class: SOA-08 Batch: A
Subject: IDS Roll No.: 08 21 Exp. No.: 2

Name of the Experiment: _____

Performed on: _____

Submitted on: _____

Marks
<u>10</u>

Teacher's Signature with date
<u>[Signature]</u>

1

Sequential Organization & Characteristics.
of linear data structures.

→

It is a method of storing data elements
in a continuous location where
elements in a continuous memory location
are arranged one after
another in a fixed order.

Characteristics:

- o elements are stored in contiguous memory locations.
- o each element has a unique index or position.
- o Data is arranged in a linear sequence.
- o Size is usually fixed.

Incomplete for Theory ☐ Diagrams ☐ Observation Table ☐
Calculations ☐ Graphs ☐ Results ☐ Conclusion ☐
Understanding Through Q/A ☐ Late Submission ☐ Neatness ☐

2
Teacher's Signature

2. Array as an abstract Data type (ADT)

ADT is collection of elements of the same data type stored in contiguous memory locations. & accessed using an index.

Components of ADT:

- Elements: Data items of the same type.
- Index: Position used to access elements.
- Size: Total number of elements.

Basic Operations on array:

- Traversal: Visiting each element.
- Insertion: Adding a new element.
- Deletion: Removing an element.
- Searching: Finding an element.

3. Insertion, deletion & Searching in an array.

Consider an array:

$A = (10, 20, 30, 40)$

insertion: insert is an position 2:

Shift elements right $\rightarrow A = (10, 20, 25, 30, 40)$

o Deletion : Delete element at position 3.

Shift elements left $\rightarrow A = (10, 20, 25, 40)$

o Searching : To search 25

Compare sequentially until found.

25 is found at index 2.

4 Advantages & limitations of arrays

Advantages:

- o fast direct access using index (O(1))
- o Simple & easy to implement
- o efficiently memory usage for fixed size data.
- o good for storing homogeneous data.

Limitations:

- o fixed size
- o Insertion & deletion are time-consuming
- o wastage of memory if size not useful

efficiently comparison:

- o Insertion : $O(n)$ due to shifting
- o Deletion : $O(n)$ due to shifting
- o Searching : $O(n)$ for linear search.
 $O(\log n)$ for binary search (sorted array).

5] 2-Dimensional array for student academic data:

2D array: Rows represent students
Columns represent marks / subject

Ex:-

Student	Math	Science	English
S ₁	85	90	88
S ₂	78	82	80

Accessing data:

- o Marks [0] [1] → Science Marks
- o traversal uses nested loops

efficient manipulations.

- o Direct indexing allows fast access.
- o Nested loop helps in processing total & Averages.

Extension to n -dimensional arrays:

- o image Processing (3D arrays for RGB)
- o Scientific Simulations
- o ML & data analysis
- o game development

